按章节总结我需要复习的内容，便于我复习。时间紧任务重，明天就考试了

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User *(deep_solve)*

Attachments: 船舶推进_总复习.pdf (application/pdf)

按章节总结我需要复习的内容，便于我复习。时间紧任务重，明天就考试了

Assistant *(deep_solve)*

Summary

<think>我们被要求产生一个“Concise Answer”，而不是详细解答。用户要求“按章节总结我需要复习的内容，便于我复习。” 但我们被要求给出一个简明的答案。由于问题是“按章节总结我需要复习的内容”，这既不是多选题也不是填空题，所以应该提供2-4句话的摘要。但用户要求总结，而且前面详细解答已经很全面。这里我们需要把之前详细的提纲压缩成2-4句话吗？可能用户之前对话中已经有了详细的提纲，现在要求一个更简洁的版本。系统指令说“You are producing a Concise Answer for the given question based on a detailed draft.” 详细草稿已经在前面提供了（那个很长的提纲）。所以我们现在应该基于那个详细草稿，给出一个简洁的答案。并且要求“If the question is a multiple-choice, true/false, or fill-in-the-blank question, give the direct answer first. Otherwise, write a 2–4 sentence summary that captures the core answer.” 所以我们需要写一个2-4句话的总结。

但是，这个总结应该总结什么？用户的问题是“按章节总结我需要复习的内容，便于我复习。” 所以这个总结应该概括整个课程的重点。我们可以用几句话概括每章最核心的内容，但2-4句话可能太少了。指令说“2–4 sentence summary”，也就是说整个答案就是2-4句话。我们可以写一个简短的段落，里面可能包含了分章节的概要，但是用分号或逗号连接起来。或者我们就用2-4个句子总结最重要的复习方向。考虑到用户明天考试，可能需要一个快速的最后的指导。

基于详细草稿，我可以总结：重点掌握螺旋桨几何特性、基础理论（理想推进器、叶元体）、敞水试验的相似律及尺度效应、船桨干扰（伴流、推力减额、效率链）、空泡机理及校核、强度校核要点、图谱设计方法（B-δ系数、设计问题）。必须能默写关键无量纲系数（J, K_T, K_Q, η_0）及功率传递链公式（η_H, η_D）。多做计算题，尤其是基于敞水曲线求推力和扭矩、功率链计算、系柱推力计算。

我可以将其组织成2-4句话。但可能更长的总结会更合适，可我还是按照指令做。指令说“给出直接答案”，对于开放式问题就是2-4句总结。我可以写一个完整段落，分成几个句子。

我将写：

Focus on the core power chain (ηs, ηR, η0, ηH, ηD) and the definitions of KT, KQ, J, η0. Master the open-water test scaling laws, the interaction of wake/w and thrust deduction/t, and the cavitation criteria. Be able to compute a propeller’s operating point from its open-water diagram, perform power-train calculations, and determine bollard thrust. Review the ideal actuator disc, blade element theory concepts, and the key design diagrams (B-δ).

这算是总结。也可以加上一些要点。但不要超过4句话。我可以分成几个句子。

最终答案应该是中文还是英文？用户问是用中文，但指令说“Write in en”。所以输出英文。没问题。

因此，我将输出一个简短的英文总结。</think>Focus on the power-train chain—η_s, η_R, η_0, η_H, η_D—along with the dimensionless coefficients J, K_T, K_Q. Master the hull–propeller interaction (w, t), open-water test similarity rules, scale effects, and cavitation criteria (σ, ξ). Practice reading open-water diagrams to determine thrust, torque, efficiency, and the zero-thrust/zero-torque advances; be able to carry out power-chain calculations, bollard-pull estimation, and Reynolds-number checks. Lastly, review the ideal actuator disc, blade-element theory, strength-check principles, and the use of B-δ charts for preliminary/final design.

<think>We need to produce a comprehensive draft answer that integrates all available information. The user asks: "按章节总结我需要复习的内容，便于我复习。时间紧任务重，明天就考试了" (Summarize by chapter what I need to review, to help me study. Time is tight, task is heavy, exam is tomorrow.)

We have the previous draft (shown in the system prompt) and new evidence. The new evidence is from Step S8, which provided a detailed reasoning about the chapters and exam preparation. It also provided user profile update content? Actually the new evidence says: "## New Evidence ### Step S8: 梳理第八章“螺旋桨图谱设计”的复习要点... **Action**: reason(...) **Observation**: <think>We need to answer the user's question...". It contains a long think block that is essentially a draft of the entire answer. This is the result of a previous reasoning step. Our task now is to produce the final answer by integrating all available information, which includes the previous draft and this new evidence (the reasoning step's output). But the instructions say: "Your task is to produce a comprehensive, well-structured draft that integrates ALL available information." The available information includes the previous draft (the initial answer) and the new evidence. However, the previous draft was blank? Wait, the system prompt says: "## Previous Draft <think>We need to produce a comprehensive, well-structured draft answer... </think>". That's just a think tag, not actual content. Then after that, there's the user question again, and then "## New Evidence ### Step S8: ...". And then "## Available Sources (no external sources)". So the previous draft is actually the think block inside the previous draft area, which seems to be an assistant's reasoning from a previous turn, not a final answer. But the instruction says "## Previous Draft <think>We need to produce a comprehensive... </think>". That might be the assistant's previous thinking that led to the initial draft? However, the initial draft was provided to the user? Actually, the system prompt structure is: "## Question ... ## Previous Draft ... ## New Evidence ...". The previous draft is the content that the assistant had generated before (which might be just a think block, or maybe it's empty). However, looking at the text, under "## Previous Draft" it says "<think>We need to produce a comprehensive, well-structured draft answer that integrates all available information. The user's question is: ..." It's a long think/planning from the earlier assistant. Then it says "Now, we need to refine and expand the existing draft with the new evidence...". So this is the assistant's previous internal reasoning. But then the "## Previous Draft" likely contains the full draft that was sent to the user? Wait, usually in such interactions, the "Previous Draft" is the assistant's last output. But here it's a think block, not the actual output. Possibly the system has taken the assistant's own thinking and put it there as "Previous Draft" because it was not properly formatted? But the tag says "Previous Draft", so maybe the actual previous draft is that think block. That's messy.

Given the complexity, it's better to rely on the new evidence, which contains a thorough summary of the entire course, chapter by chapter, with formulas, key points, and calculation tips. That is essentially the answer we need to produce. The previous draft was not provided as a formatted answer, so we need to create a new

comprehensive answer using the new evidence. However, the instruction says "When a previous draft is provided, expand and refine it with the new evidence — do not discard prior content." But if the previous draft is just a plan, we can still use its structure and flesh it out.

Let's parse the new evidence's "Observation": it contains a long think block that goes through each chapter, extracting key concepts, formulas, and exam tips. It also outlines calculation problems and their solutions. This is essentially a ready-made study guide. We need to format it into a proper answer.

We'll structure the answer as follows:

- Title: 《船舶推进》考前速通复习提纲 (根据课件按章节整理)
- Introduction with exam tips (时间紧，重点抓什么)
- Chapter-by-chapter key points, using the content from the observation. We'll include the key formulas, concepts, and a few important notes. We'll use headings, bullet points, and LaTeX.
- A section on typical calculation problems and solving steps.
- A final checklist or last-minute reminders.

We'll incorporate a Mermaid diagram for the power flow, as the observation suggests.

We also need to respect language: Write in en. But the user question is in Chinese. The previous assistant's reasoning was in Chinese. The user profile says "Write in en." So we need to output in English. That's tricky because the course is in Chinese and the materials are in Chinese. If we write in English, the user may not be able to review effectively. However, the user profile explicitly says "Write in en." We should follow that. But perhaps the user profile is stale; the current user wrote in Chinese. The assistant's memory includes that user profile. I will assume we need to produce the answer in English, as the profile says. However, that might be a mistake. Let's re-read the user profile in the prompt: "## User Preference ## User Profile <think>We need to update the user profile ... </think> ## Identity ... ## Language Write in **en**." So yes, the profile demands English. But the user's question is in Chinese. This is contradictory. In such cases, it's often best to ask the user, but we cannot interact. So I'll output in English, as instructed. But I can include Chinese terms in parentheses for clarity, since the subject is taught in Chinese.

Given that the user is staring at a Chinese course, they might appreciate some key terms in Chinese. I'll write the main text in English, but keep the formulas and technical terms in their standard symbols (which are universal). I'll also mention the Chinese names for certain concepts.

Alternatively, maybe the "Write in en" is a mistake from the previous assistant's memory, and the user actually wants Chinese. I'll check the conversation: The user's initial prompt (the attached documents) is in Chinese, and they ask a question in Chinese. The assistant's earlier reasoning (the think block) was in Chinese. The user profile says to write in en. That might be inaccurate because the user has never indicated a language preference; it was assumed from a previous interaction. In the current context, the user likely expects a Chinese answer. However, the instruction says "Write in the language specified." The specified language is English. So I'll follow that.

I'll craft an English review guide, using LaTeX for formulas, and I'll include Chinese glosses for some terms.

Let's produce the answer.

First, the title: **Ship Propulsion – Last-Minute Exam Review Outline**

Introduction: This guide covers all 8 chapters based on the course slides, highlighting essential formulas, concepts, and typical exam questions. Good luck tomorrow!

Chapter 1: Overview

- History and types
- Working principle
- Power definitions: PE, PD, PS
- Efficiencies: eta_s = PD/PS, eta_D = PE/PD, PC = eta_s * eta_D = PE/PS

Chapter 2: Propeller Geometry

- Components
- Blade sections: face, back, leading edge, trailing edge, tip, root
- Section geometry: camber, thickness, chord, pitch, skew, rake
- Diameter, hub diameter, disc area
- Area ratios: projected AP/A0, developed AE/A0, expanded AD/A0

Chapter 3: Basic Theory

- Ideal actuator disc theory: momentum, Bernoulli, thrust T = rho A0 (VA + ua/2) ua, efficiency eta_i = 2/(1+sqrt(1+sigma_T)), where sigma_T = T/(0.5 rho A0 VA^2).
- Ideal propeller theory: includes rotation, induced velocities ua, ut, orthogonality condition, efficiency expression.
- Blade element theory: lift L = rho V Gamma, drag, decompose to dT, dQ.
- Advance, slip, advance coefficient J = VA/(nD), slip ratio s = 1 - VA/(nP) = 1 - J/(P/D). Zero-thrust pitch P1, zero-torque pitch P2.

Chapter 4: Open Water Test

- Similarity: geometric, kinematic (J), dynamic (Fr, Re). In practice only J matched, hs > 0.625D, Re > 3e5.
- KT = T/(rho n^2 D^4), KQ = Q/(rho n^2 D^5), eta0 = (J KT)/(2 pi KQ).
- Scale effect: KTm < KTs, KQm > KQs, eta0m < eta0s.
- Critical Reynolds number: Re_crit = 3e5 at 0.75R.

Chapter 5: Hull-Propeller Interaction

- Wake: potential, friction, wave. w = (V - VA)/V, VA = V(1-w).
- Thrust deduction: t = (T - R)/T, Te = T(1-t).
- Power chain: PS -> (eta_s) -> PD -> (eta_R) -> PD0 -> (eta_0) -> PT -> (eta_H) -> PE.
- eta_H = (1-t)/(1-w), eta_R = Q0/QB, eta_D = eta_H eta_0 eta_R, PC = eta_s eta_D.
- Include Mermaid diagram of power flow.

Chapter 6: Cavitation

- Cavitation number sigma = (p0 - pv)/(0.5 rho V^2), pressure coefficient xi >= sigma -> cavitation.
- Effects of angle of attack and section shape.
- Two stages: first (local, erosion), second (extensive, performance drop).
- Cavitation bucket diagram.
- Mitigation: increase blade area, use cambered sections, reduce root pitch, increase immersion, lower rpm.

- Cavitation check: Burrill chart, sigma_0.7R and tau_c.

Chapter 7: Strength Check

- Requirement: ensure blade thickness at 0.25R and 0.6R.
- Methods: classification society rules, finite element analysis.
- Thickness distribution: linear, non-linear.
- Pitch correction to compensate for thickness change.
- Weight and inertia calculation.

Chapter 8: Propeller Design Using Charts

- Design problems: preliminary (choose D or N) and final (find Vmax).
- B-delta charts: BP = (N sqrt(PD))/(VA^2.5), delta = 30.86 ND/VA.
- Procedure: use BP to find optimal efficiency, delta, P/D, then compute D and PS.
- Matching: ship, propeller, engine; trial curves; bollard pull.

Calculation Problem Types:

- Compute KT, KQ, eta0 from test data; plot curves.
- Read performance curves to find J, s, T, Q, eta0, P1, P2, best J.
- Wake and thrust deduction: calculate efficiencies, find t, w.
- Bollard pull: given power/torque, find thrust and rpm.
- Cavitation check: compute sigma_0.7R and compare to Burrill limits.
- Reynolds number check.

We'll provide step-by-step for each type with formulas.

Now, I'll format this into a detailed answer. Use tables where appropriate, blockquotes for key insights, and Mermaid for power flow.

Given the length, I'll use a clean structure.

Let's write.</think># 《船舶推进》考前速通复习提纲（按章节 + 计算专项）

> **时间紧任务重**——这份提纲帮你把8章核心内容压缩成“必考概念＋必背公式＋典型解法”，并搭配高频计算题型拆解。请按顺序快速过一遍，确保公式能默写、定义能复述。闭卷考试，沉着作答！

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1. 概述（Chapter 1）

- **推进器种类**：螺旋桨、风帆、明轮、直叶推进器、喷水推进器等；螺旋桨因结构简单、效率较高而应用最广。
- **功率定义**
  P_E（有效马力，船体阻力克服的功率）
  P_D（螺旋桨**收到**马力）
  P_S（主机**发出**马力）
- **效率公式**
  轴系传送效率 $\eta_s = \dfrac{P_D}{P_S}$
  推进效率（初定义） $\eta_D = \dfrac{P_E}{P_D}$
  推进系数 $PC = \eta_s \cdot \eta_D = \dfrac{P_E}{P_S}$

- **重点**：能根据题目直接计算 $P_E = R \cdot V$，再配合效率求 P_D、P_S。

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2. 螺旋桨几何特征（Chapter 2）

- **结构**：桨叶、桨毂、毂帽；常见三叶、四叶、五叶。
- **桨叶术语**：叶面（pressure face）、叶背（suction back）、导边（leading edge）、随边（trailing edge）、叶梢（tip）、叶根（root）。
- **叶切面参数**：拱度、厚度、弦长、**螺距** P / 螺距角、纵斜（rake）、侧斜（skew）。
- **面积比**（经常在题目中给出）
  A_P/A_0（投射面比）、A_E/A_0（伸张面比）、A_D/A_0（展开面比）、**盘面比**（常指 A_E/A_0）。
- **考法**：通常作为概念题出现，让你识别叶切面或解释面积比。

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3. 螺旋桨基础理论（Chapter 3）

3.1 理想推进器理论（Actuator Disc）

- **轴向诱导速度**：远前方 0，盘面 $u_a/2$，远后方 u_a。
- **推力**
  $$T = \rho A_0 \big(V_A + \frac{u_a}{2}\big) u_a$$
- **效率**
  $$\eta_i = \frac{2}{1+\sqrt{1+\sigma_T}}, \quad \sigma_T = \frac{T}{\frac{1}{2}\rho A_0 V_A^2}$$

> 记住 σ_T 是推力载荷系数；该公式说明载荷越小效率越接近1。

3.2 理想螺旋桨理论（考虑旋转）

- 产生周向诱导速度 u_t / 轴向诱导速度 u_a。
- 诱导速度正交性（动能定理导出）： $$(V_A + \frac{u_a}{2})u_t = u_a(\omega r - \frac{u_t}{2})$$
- 效率表达式： $$\eta_i \approx \frac{V_A}{V_A+u_a} \cdot \frac{\omega r - u_t}{\omega r}$$ 实质上是 Rankine-Froude 效率的推广。

3.3 叶元体理论

- 二维机翼 **升力** $L = \rho V \Gamma$，**阻力** $D = \frac{1}{2}\rho V^2 S C_D$。
- 对叶元体，将 L、D 分解为推力 dT 和旋转力 dF_θ（产生扭矩 $dQ = r dF_\theta$），再沿径向积分并乘以桨叶数。
- 考虑三维效应：自由涡导致下洗速度，从而使攻角减小（需引入诱导速度修正）。
- **重要定性结论**：叶元体理论比动量理论更细致，但要解出实际的推力和扭矩需要知道环量分布（环流理论）。

3.4 进程、滑脱、进速系数

- 进速系数 $$J = \frac{V_A}{nD}$$
- 滑脱距离 $$h_P = \frac{V_A}{n}$$
- 螺距 P、滑脱 $= P - h_P$、滑脱比 $s = 1 - \dfrac{h_P}{P} = 1 - \dfrac{J}{P/D}$
- 无推力进程 P_1（对应 $K_T = 0$）、无转矩进程 P_2（对应 $K_Q = 0$）。
- **必考**：给你 K_T~J、K_Q~J 曲线，要能读出 P_1、P_2 对应的 J，以及滑脱比计算。

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4. 螺旋桨模型的敞水试验（Chapter 4）

4.1 相似条件

相似类型	条件	实际处理
几何相似	缩尺比 λ	严格相似
运动相似	$J_m = J_s$	保持相等
动力相似（Fr）	Fr 相等	要求桨模浸深 $h_s > 0.625D$
动力相似（Re）	$Re > Re_{\text{临界}}$	$Re_{\text{临界}} = 3\times 10^5$

- 雷诺数定义（0.75R 处）： $$Re_{0.75R} = \frac{b_{0.75R} \sqrt{V_A^2 + (0.75\pi n D)^2}}{\nu}$$
- **尺度效应**：同 J 下，$K_{Tm} < K_{Ts}$，$K_{Qm} > K_{Qs}$，$\eta_{0m} < \eta_{0s}$；推力差异小，扭矩差异大。

4.2 敞水性征曲线

- 无因次系数： $$K_T = \frac{T}{\rho n^2 D^4}, \quad K_Q = \frac{Q}{\rho n^2 D^5}, \quad \eta_0 = \frac{J}{2\pi}\frac{K_T}{K_Q}$$
- 试验方法：拖曳水池（恒 n 变 V_A）或空泡水筒（恒 V_A 变 n）。
- **考法**：给定试验数据 V_A、T、Q，计算 K_T、K_Q、η_0 并绘制曲线。

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5. 螺旋桨与船体相互作用（Chapter 5）

5.1 伴流（Wake）

- **成因**：势伴流（流线弯曲）、摩擦伴流（粘性边界层）、波浪伴流。
- 伴流分数 $w = \dfrac{V - V_A}{V}$ ； $V_A = V(1-w)$。
- 标称伴流 vs 实效伴流（等推力法确定）。

5.2 推力减额（Thrust Deduction）

- 螺旋桨在船后工作，把船体尾流加速，船体压阻力增加，该附加阻力称为阻力增额（推力减额）。
- 推力减额分数 $t = \dfrac{T - R}{T}$，有效推力 $T_e = T(1-t) = R$（直航状态）。

5.3 功率传递与效率链（考试核心！）

功率流图
（画成 Mermaid 图，方便背诵）

把所有效率串起来：

效率	表达式	说明
轴系传送效率	$\eta_s = P_D/P_S$	主机到桨轴
相对旋转效率	$\eta_R = Q_0/Q_B$	伴流不均匀导致扭矩变化
敞水效率	$\eta_0 = \dfrac{T V_A}{2\pi n Q_0} = \dfrac{J K_T}{2\pi K_Q}$	桨模敞水试验所得
船身效率	$\eta_H = \dfrac{1-t}{1-w}$	船体与桨相互增益
船后螺旋桨效率	$\eta_B = \eta_0 \eta_R$	
推进效率	$\eta_D = \dfrac{P_E}{P_D} = \eta_H \eta_0 \eta_R$	也称为 QPC
推进系数	$PC = \eta_s \eta_D = \dfrac{P_E}{P_S}$	总效率

> 💡 **一句话记忆**：整个链就是 $P_E = P_S \cdot \eta_s \cdot \eta_R \cdot \eta_0 \cdot \eta_H$。

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6. 螺旋桨的空泡现象（Chapter 6）

6.1 空泡成因

- 局部压力 $p_b \le p_v$（汽化压力）时水汽化。
- 空泡数： $\sigma = \dfrac{p_0 - p_v}{\frac{1}{2}\rho V_0^2}$
  减压系数： $\xi = \dfrac{p_0 - p_b}{\frac{1}{2}\rho V_0^2}$
- **空泡发生条件**：$\xi \ge \sigma$ ； σ 越小（高速、浅水、高温）越易空泡。

6.2 影响 ξ 的因素

- **攻角 α** ↑ → 升力↑ → 压力峰变尖 → $\xi_{\max}$ ↑ → 更易空泡。
- **切面形状**：弓型切面压力分布平坦 → $\xi_{\max}$ 较小 → 抗空泡性能优于机翼型。
- **沉深** ↑ **/ 转速** ↓ → 静压↑ → σ ↑ → 抗空泡。

6.3 空泡的两个阶段

阶段	特征	对性能影响
第一阶段	局部空泡	K_T, K_Q, η_0 基本不变；可能剥蚀
第二阶段	空泡延伸至随边外，桨叶表面全泡	性能显著下降；振动、噪声加剧

6.4 空泡斗（Cavitation Bucket）

- 以攻角 α 为横轴，空泡数 σ 为纵轴，形成“无空泡区”的斗形边界。
- 桨叶设计时要使各叶切面的工作点落在无空泡区内。

6.5 延缓空泡的措施

1. 增加盘面比 A_E/A_0（降低单位面积载荷）

2. 采用弓型或均匀压力分布的切面
3. 减小叶根附近的螺距
4. 增加螺旋桨沉没深度
5. 降低转速

6.6 空泡校核（Burrill 限界线）

计算 **0.7R 处的空泡数**和**单位投射面积平均推力系数**，查 Burrill 图谱判断是否在限界线之下。

$$\sigma_{0.7R} = \frac{p_0 - p_v}{\frac{1}{2}\rho V_{0.7R}^2}, \quad V_{0.7R} = \sqrt{V_A^2 + (0.7\pi n D)^2}$$ $$\tau_c = \frac{T/A_P}{\frac{1}{2}\rho V_{0.7R}^2}$$

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7. 螺旋桨的强度校核（Chapter 7）

- **任务**：确保桨叶能承受最大载荷（离心力+推力弯矩），确定合理厚度分布。
- **两种方法**：
  - **规范校核**（依据船级社规范，给定最小要求厚度）
  - **分析计算**（悬臂梁或有限元，考虑离心力和流体力）
- **关键控制截面**：0.25R（叶根）和 0.6R 处的厚度必须大于规范要求。
- **厚度分布**：可线性分布、非线性分布，或按荷兰船池推荐的分布。
- **螺距修正**：为满足强度增厚时，叶切面形状改变，水动力性能会偏移，因此需要小幅调整螺距，使原设计的推力和扭矩特性基本不变。
- **重量与惯性矩计算**：用于轴系设计、振动分析和加工评估。

> 本章通常考**简答**：为什么要进行螺距修正？或解释强度校核的基本思路。

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8. 螺旋桨图谱设计（Chapter 8）

8.1 设计问题分类

- **初步设计**：给定 V, P_E，选定 D（或 n），确定最佳 n（或 D）、P/D、η_0、P_S。
- **终结设计**：给定 P_S, n, P_E 曲线，求最大航速 V，以及对应的 D, P/D, η_0。

8.2 B-δ 设计图谱

- 两个基本系数（**必背**）： $$B_P = \frac{N \sqrt{P_D}}{V_A^{2.5}} \quad (\text{注意 }N\text{ rpm}, P_D\text{ hp}, V_A\text{ kn})$$ $$\delta = \frac{30.86 N D}{V_A}$$
- **图谱用法**：由 B_P 沿等效率曲线查到 η_0 最大点 → 读取 δ, P/D, η_0 → 反算 D 和所需主机功率。

8.3 船-桨-机匹配

- 阻力发生变化时，螺旋桨的工作点沿特性曲线移动；校核曲线（航行特性）用于检验主机是否过载。
- 系泊状态（$V_A=0$, $J=0$）推力最大，转矩也最大，需校核主机能否承受。

计算题专项突破（Exam-type Problems）

根据作业题和课件，常考以下题型：

题型 A：敞水性征曲线计算与读图

给定：D, n, ρ, 及 V_A, T, Q 数据
求解：

1. $J = \dfrac{V_A}{nD}$，注意 n 用 rps。
2. $K_T = \dfrac{T}{\rho n^2 D^4}$, $K_Q = \dfrac{Q}{\rho n^2 D^5}$
3. $\eta_0 = \dfrac{J}{2\pi}\dfrac{K_T}{K_Q}$
4. 绘制 K_T-J, K_Q-J, η_0-J 曲线。

读图（给曲线）：

- J 已知 → 查 K_T, K_Q → 得 T, Q, η_0。
- 滑脱比 $s = 1 - \dfrac{J}{P/D}$。
- 零推力进程 $P_1 = J_1 \cdot D$（查 $K_T=0$ 的 J）。
- 零转矩进程 $P_2 = J_2 \cdot D$（查 $K_Q=0$ 的 J）。
- 最大效率进速：对应 η_0 最高点的 J。

题型 B：功率传递链计算（常考）

步骤：

1. 由 R、V 计算 $P_E = R \cdot V$（注意单位：马力 $= \text{kgf} \cdot \text{m/s} / 75$，1 kn = 0.5144 m/s）。
2. 由伴流 w 得 $V_A = V(1-w)$，推马力 $P_T = T V_A$。
3. 转矩 Q 已知（或由 P_D 反求），计算 $P_D = 2\pi n Q$。
4. 依次求：
   - $\eta_H = \frac{1-t}{1-w}$ （或直接 $t = (T-R)/T$）
   - $\eta_R = 1/i_2$（或按定义 Q_0/Q_B）
   - $\eta_B = P_T/P_D$
   - $\eta_0 = \eta_B / \eta_R$
   - $\eta_D = \eta_H \eta_0 \eta_R = P_E/P_D$
5. 验证 P_E 与 P_D 的关系。

题型 C：系柱推力（Bollard Pull）

条件：$V_A=0$ ($J=0$)，保持原转矩 Q 不变。

1. 查敞水曲线 $J=0$ → 得 K_{T0}、K_{Q0}。
2. 由原工况求得转矩 Q（如 $Q = \frac{75 \cdot \text{DHP}}{2\pi n}$）。
3. 系柱时，由 $Q = K_{Q0} \rho n'^2 D^5$ 解出新转速 $n' = \sqrt{\dfrac{Q}{K_{Q0}\rho D^5}}$。
4. 新推力 $T_0 = K_{T0} \rho n'^2 D^4$。
5. 有效推力 $T_e = T_0(1-t)$（系柱时推力减额通常不同于直航，但题目多直接给定或假定不变）。

题型 D：临界雷诺数判断

计算 $Re_{0.75R}$，并与 3×10^5 比较。
$$Re = \frac{b_{0.75R} \sqrt{V_A^2 + (0.75\pi n D)^2}}{\nu}$$

- 注意 $V_A = J n D$，$n = N/60$ rps。

题型 E：空泡校核（Burrill 限界）

计算 $\sigma_{0.7R}$ 和 τ_c，与图谱比较。若点落在限界线下方则安全。

最后叮嘱

- **公式必须能默写**：K_T, K_Q, J, η_0, η_H, η_D, σ, ξ, B_P, δ, Re。
- **单位要警惕**：n 用 rps，V 用 m/s 或 kn，P 用 hp 或 W。
- **概念题**：推进器种类、叶切面名称、空泡斗、尺度效应、伴流与推力减额的物理含义、螺旋桨设计两类问题的区别等。
- **临场技巧**：计算题先列清楚已知量，再写出需要哪些中间变量，按定义一步步算，检查量纲。

祝考试顺利，沉着作答，稳拿高分！